[METRIC]
A-A-3149
January 28, 1998
*Superseding*
Fed. Spec. O-P-1731
October 5, 1979 and
Int. Amend. -1
May 7, 1979

# COMMERCIAL ITEM DESCRIPTION

## PENETRATING FLUID

The General Services Administration has authorized the use of this commercial item description as a replacement for Federal specification O-P-1731, for all federal agencies.

1. Scope. *This commercial item description covers a liquid penetrating fluid, for loosening frozen* metallic parts, and three classes of containers.

2. Types and classes.

Class 1 - non aerosol container
Class 2 - aerosol container, with non-flammable propellant
Class 3 - aerosol container, with flammable propellant

3. Salient Characteristics:

3.1 Material. The penetrating fluid shall be formulated as necessary to meet the requirements of this commercial item description. The penetrating fluid shall be non-flammable and contain no mineral oil.

3.2 Propellant.

3.2.1 Class 2 (aerosol) penetrating fluid (non-flammable) propellant shall be composed of carbon dioxide, nitrous oxide, or nitrogen, and shall be non-flammable.

3.2.2 Class 3 (aerosol penetrating fluid (flammable) propellant shall be composed of a mixture of propane and isobutane.

3.3 Toxicity. When used for its intended purpose, the penetrating fluid and propellant, when applicable, shall have no adverse effect on the health of personnel.

4. Physical properties. The penetrating fluid shall conform to requirements shown in Table I and paragraph 4.1.

Beneficial comments, recommendations, additions, deletions, clarification, etc., and any data which may improve this document, should be sent to: GSA, Paints and Chemicals Center, 400 15th Street, S.W., Auburn, WA 98001.

FSC 6850

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TABLE I. Physical properties

| Characteristic | Min | Max | ASTM Method |
|---|---|---|---|
| pH at 25$^0$C, (77$^0$F) | 9.0 | 9.7 | E70 |
| Pour Point, | — | -3$^0$C(27$^0$F) | D97 |
| Refractive index, at 20$^0$C (68$^0$F) | 1.3516 | 1.3636 | D1218 |
| Viscosity, Centistokes at 38$^0$C (100$^0$F) | — | 1.5 | D445 |
| Flash point | None to boiling | | |
| Surface tension, dynes per centimeter | — | 35.0 | D1331 |

4.1 Solubility. Use equal portions of penetrating fluid and distilled water to form a composite specimen of not less than one pint. Maintain the solution at 25$^0$C +/- 2$^0$C (77$^0$F +/- 2$^0$F), and stir vigorously for 5 minutes. Examine visually. There shall be no trace of separation of the penetration fluid, milkiness, or turbidity in the specimen.

4.2 Aerosol containers. The product shall be packaged in an aerosol dispenser. The dispenser shall produce a spray in the form of a mist when moderate finger pressure is applied on the actuator. The dispenser shall stop the release of the spray immediately upon release of the pressure on the actuator. The dispenser shall deliver at least 98 percent by weight of its contents.

4.3 Net weight. The net weight of the contents of the aerosol dispenser shall be a minimum of 283 grams (10 ounces, avoirdupois.)

4.4 Seepage. When tested as specified in ASTM D 3094 -Seepage Rate of Aerosol of Aerosol Products, the dispensers shall have a seepage rate no greater than 3.0 ml, using wide mouth eudiometer tubes similar to that in Figure 2 of ASTM D 3094.

4.5 Container stability. The filled containers shall not leak or be damaged by internal pressure when heated to 54$^0$C (130$^0$F.)

NOTE: The contractor may use his normal quality assurance procedures for manufacture of this product; however, should a disagreement arise between the Government and its customer or between the Government and the manufacturer concerning compliance to the salient characteristics of this commercial item description, the test methods and tests stated in this commercial item description will be used to determine compliance of the product.

3. Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

ASTM standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

PREPARING ACTIVITY:
GSA - FSS



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